Fall 2016 / EC571

Instructor:Ajay Joshi

Final Exam

Date: December 17, 2016

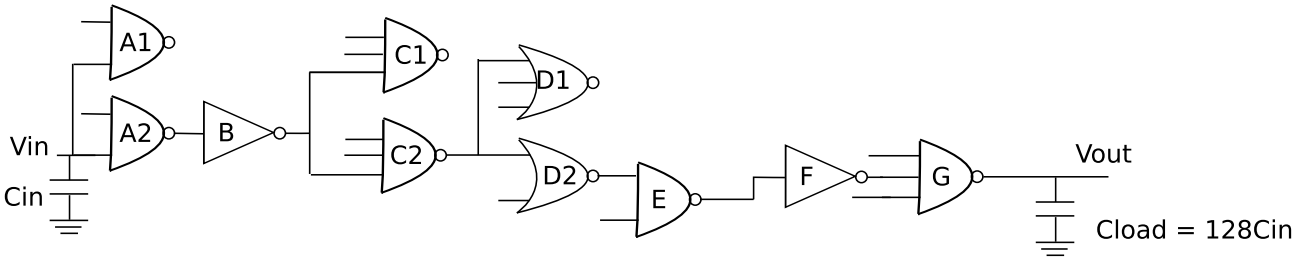
Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
7	20	
Bonus	5	
Total	100	

1. The duration of the exam is 2 hours.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. The solutions should be written in the spaces provided on the test sheet.
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
7. Material sharing is not allowed.
8. Calculators are allowed.
9. Boston University Academic Code of Conduct applies.

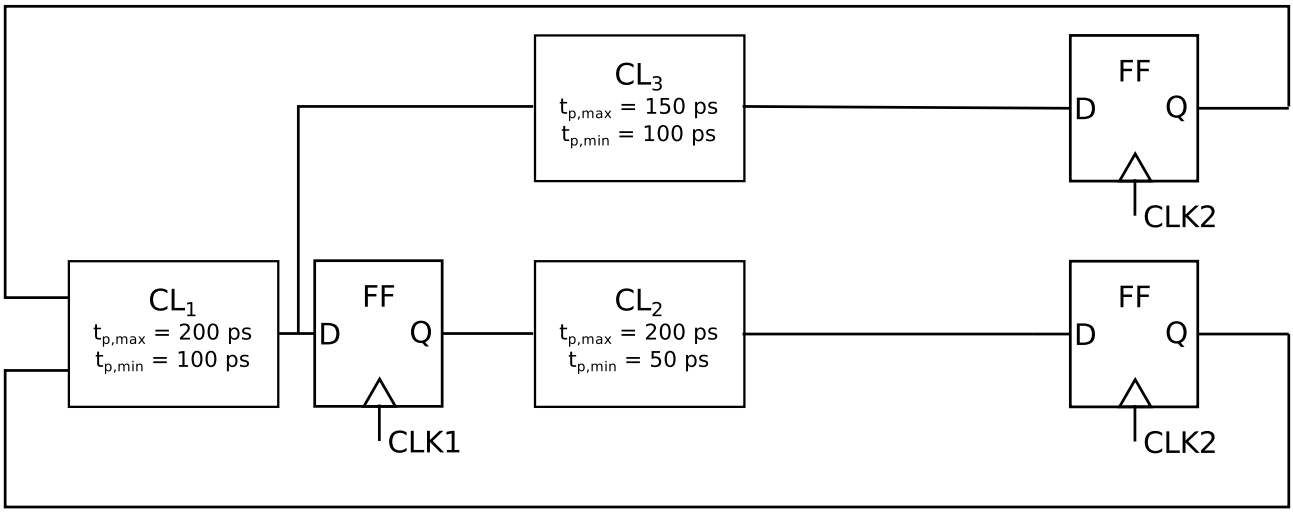
Question 1



For the circuit shown in the above figure, using the logical effort technique, calculate the input width of transistors for each gate in the path such that total delay of the path is minimized. You can assume that gates A1 and A2 are identical and C1 and C2 are identical. Assume Cin = 2 fF, Cg = 1 fF/u. (20 points)

Question 2

Consider the pipelined sequential logic block shown below. The minimum and maximum propagation delay of each combinational logic block is annotated to each combinational logic block. Assume $t_{clk-to-q} = t_{setup} = t_{hold} = 50\ ps$.



1. What is the minimum clock period for the pipelined sequential logic block above? Assume there is no clock skew or jitter. Are there any hold-time violations? (7 points)

2. Assume CLK1 and CLK2 can have a skew of $\pm$ 60 ps relative to each other. What is the minimum clock period in that case? Are there any hold-time violations? (8 points)

3. Explain the difference between a flip-flop and a latch. (5 points)

Question 3

Sketch the schematics for $f(A,B,C) = \overline{A \cdot (B+C)}$ using each of the following circuit techniques. You can assume that you have both complemented and non-complemented inputs:

1. Static CMOS. (5 points)

2. Pseudo-nMOS. (5 points)

3. Dynamic. (5 points)

4. Pass transistors (including the inverter at the end). (5 points)

Question 4

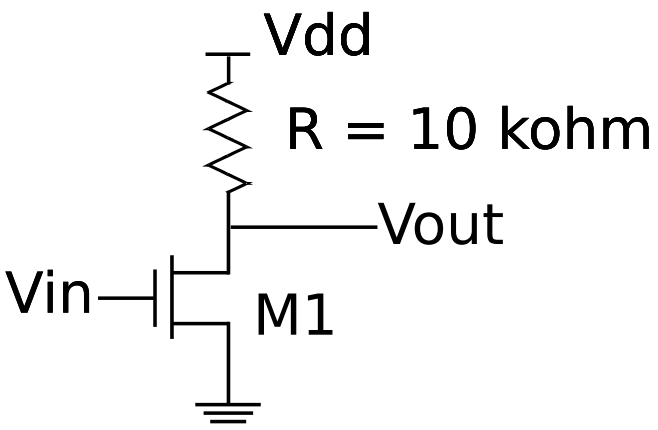


Figure 1

Assume V_{dd} = 2.5 V, $V_{TH,n}$ = 0.5 V, W_{M1} = 500 nm, L_{M1} = 100 nm, μ_n = 400 $cm^2/(V.s)$, C_{ox} = 1.125 $\mu F/cm^2$ and no short-channel effects.

1. For V_{in} = 0 V, what is the value of V_{out}? (2 points)

2. For V_{in} = 2.5 V and V_{out} = 2.5 V, what is the region of operation for M1? (2 points)

3. Write the equation for calculating I_D in that region. (2 points)

4. Calculate the value of I_D. (2 points)

5. With V_{in} fixed at 2.5 V, V_{out} will start reducing. At what output voltage does the region of operation change for M1? (2 points)

6. Write the equation for calculating I_D for this new region of operation. (2 points)

7. Calculate the value of I_D. (2 points)

8. As V_{out} further decreases, what happens when $V_{out} << (V_{in} - V_{TH,n})$? (2 points)

9. Write the equation for calculating I_D for this new region of operation. (2 points)

10. Calculate the value of I_D. (2 points)

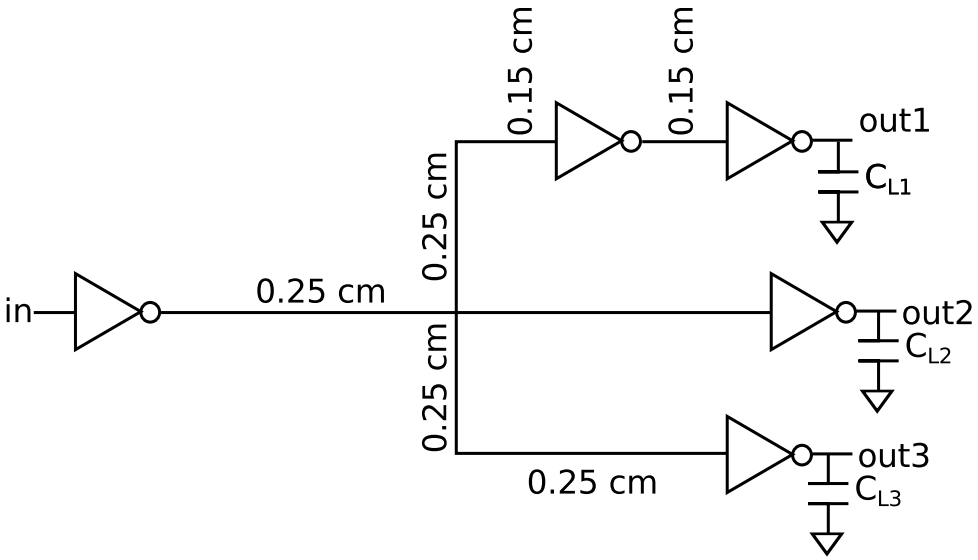
Question 5

Design the static complementary CMOS implementation of an encoder circuit represented by the truth table shown below. Design the circuit for minimum delay (assuming a stage effort of 4) and driving a load of 10 fF. As part of the design you need to determine the width of all transistors. You can use the following transistor parameters for the design - L = 0.25 micron, minimum transistor width = 0.3 micron, $R_{sq}(NMOS)$ = 10 Kohm/sq, $R_{sq}(PMOS)$ = 25 Kohm/sq, C_g = 1.5 fF/micron, C_d = 1.5 fF/micron. You can assume that you have both complemented and non-complemented inputs and that input signals have very small rise and fall times. *Hint: Determine the Boolean expression for each output column of the encoder circuit and use the Boolean expression to design the circuit. Use logical effort technique to determine the transistor sizes.* (20 points)

Table 1: Truth table for Encoding

Conventional Binary Encoded Value	New Encoded Value
$b_3\ b_2\ b_1\ b_0$	$e_1\ e_0$
0 0 0 0	1 1
0 0 0 1	1 0
0 0 1 0	1 X
0 0 1 1	1 0
0 1 0 0	0 0
0 1 0 1	X 0
0 1 1 0	0 X
0 1 1 1	X 0
1 0 0 0	0 X
1 0 0 1	X 1
1 0 1 0	0 X
1 0 1 1	X 1
1 1 0 0	0 0
1 1 0 1	1 X
1 1 1 0	0 0
1 1 1 1	1 1

Question 6



1. For the circuit shown in the above figure, draw the equivalent RC network that includes all the parts. Assume a lumped model for each individual segment of the interconnect. (5 points)

2. Write the Elmore delay equations for the three paths from in to out1, in to out2 and in to out3. (5 points)

3. Calculate the Elmore delay for all three paths. You can use the following technology parameters for calculating the delay - Vdd = 1.5 V, L = 0.25 micron Rsq(NMOS) = 10 Kohm/sq, Rsq(PMOS) = 25 Kohm/sq, Cg = 1 fF/micron, Cd = 1.5 fF/micron, wire capacitance = 0.25 fF/micron, wire resistance = 15 ohms/sq. The interconnect has a width of 0.5 micron. Assume that all inverters use a minimum width of 300 nm for the NMOS. You will need to choose the width of the PMOS transistor such that the low-to-high delay of the inverters matches with the high-to-low delay. Assume $C_{L1} = C_{L2} = C_{L3} = 20$ fF. (5 points)

4. Calculate the energy consumed in all three paths when the input ‘in’ changes from logic 1 to logic 0. Assumed Vdd = 1.5 V(5 points)

Bonus question (5 points)
Prof. Joshi thinks that winter is the best time to have ice cream. Name 5 different ice cream flavors he should try this winter.